# Advertising Spend Vs Sales Analysis

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**A PROJECT REPORT**

***Submitted by***

***Samarth Khare(24BCS10404)***

***in partial fulfillment for the award of the degree of***

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# BONAFIDE CERTIFICATE

Certified that this project report **"ADVERTISING SPEND VS SALES ANALAYSIS DASHBOARD"** is the bonafide work of "SAMARTH KHARE" who carried out the project work under my/our supervision.

**SIGNATURE**

**SIGNATURE**

**HEAD OF THE DEPARTMENT**

**SUPERVISOR**

**INTERNAL EXAMINER.**

**EXTERNAL EXAMINER**

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# ABSTRACT

In today's data-driven world, organizations generate large volumes of data through their daily operations, particularly in the retail and e-commerce sectors. However, the true value of this data lies in its analysis and interpretation. Seasonal variations in sales significantly influence business performance, making it essential for organizations to understand patterns and trends over time. Effective analysis of seasonal sales helps businesses improve decision-making, optimize inventory, and enhance customer satisfaction.

The primary objective of this project is to analyze seasonal sales patterns using Microsoft Power BI, a powerful business intelligence and data visualization tool. The dataset used for this project has been obtained from Kaggle and contains detailed information such as product categories, product names, regions, revenue, unit prices, total units sold, and payment methods. This dataset provides a comprehensive foundation for performing multi-dimensional analysis.

The project follows a structured approach that includes data collection, data cleaning, transformation, modelling, and visualization. Power Query Editor is used to preprocess the data by handling missing values, removing inconsistencies, and creating additional columns such as month, year, and season. Data modelling is performed by establishing relationships between different fields, and DAX (Data Analysis Expressions) is used to create measures such as total revenue and total units sold.

The final output of the project is an interactive dashboard that provides insights into monthly sales trends, seasonal performance, category-wise revenue, regional distribution, and payment methods. The analysis reveals that sales vary across seasons, with certain periods generating higher revenue. It also highlights the dominance of specific product categories and regions, as well as customer preferences in payment methods.

In conclusion, this project demonstrates the effectiveness of Power BI in transforming raw data into meaningful insights. The developed dashboard enables users to analyze sales patterns efficiently and supports data-driven decision-making. This study highlights the importance of business intelligence tools in modern organizations and provides a foundation for further advancements such as predictive analytics and real-time data analysis.

# CHAPTER 1

# INTRODUCTION

## 1.1. Identification of Client /Need / Relevant Contemporary issue

In the current era of digitalization, businesses are increasingly relying on data to drive strategic decisions and improve operational efficiency. The retail and e-commerce sectors, in particular, generate vast amounts of transactional data on a daily basis. However, despite the availability of such large datasets, many organizations struggle to effectively analyze and interpret this data, especially when it comes to understanding seasonal variations in sales. This creates a significant gap between data availability and actionable insights.

The issue of ineffective sales analysis is well documented in various industry reports and studies. According to multiple retail analytics studies, businesses can lose up to 20–30% of potential revenue due to poor demand forecasting and lack of data-driven strategies. Seasonal fluctuations, such as increased demand during festive periods or reduced sales during off-seasons , further complicate the problem. Without proper analysis, organizations may either overstock or understock products, leading to financial losses and reduced customer satisfaction.

From a client perspective, the problem is highly relevant and requires immediate attention. The client in this scenario can be considered as a retail or e-commerce organization that seeks to improve its sales performance and decision-making process. Such organizations require a system that can provide clear insights into sales trends, identify high-performing products, and highlight seasonal patterns. Currently, many businesses rely on traditional tools such as spreadsheets, which are not capable of handling large datasets efficiently or providing interactive visualizations. This creates a need for a more advanced and user-friendly solution.

The need for seasonal sales analysis is further supported by survey-based findings. Various surveys conducted in the retail sector indicate that a majority of businesses face challenges in understanding customer demand patterns. For instance, reports suggest that over 60% of small and medium enterprises

(SMEs) do not fully utilize their data for strategic decision-making. This highlights a critical need for tools that can simplify data analysis and present insights in an easily understandable format.

Additionally, the relevance of this issue is evident in contemporary technological trends. Leading organizations and research agencies emphasize the growing importance of Business Intelligence (BI) tools in modern enterprises. Reports from technology firms highlight that the adoption of BI tools like Microsoft Power BI has significantly increased in recent years due to their ability to provide real-time insights and interactive dashboards. These tools help organizations transform raw data into meaningful information, enabling better planning and forecasting. Therefore, the problem of ineffective seasonal sales analysis is not only real but also highly significant in today's business environment. It affects organizational performance, decision-making, and overall profitability. Addressing this issue through the implementation of an Efficient data visualization and analysis system is essential for businesses to remain competitive. This project aims to fullfill this need by developing a Power BI-based dashboard that provides comprehensive insights into seasonal sales patterns.

## 1.2. Identification of Problem

In the modern business environment, organizations generate a vast amount of data through daily transactions, especially in the retail and e-commerce sectors. Despite the availability of such data, many organizations face significant challenges in effectively analyzing and interpreting it. One of the major issues is the inability to identify and understand patterns in sales data over time, particularly those related to seasonal variations. This lack of understanding creates a gap between data availability and meaningful insights. Seasonal fluctuations in sales are influenced by multiple factors such as festivals, holidays, weather conditions, and changing customer preferences. These variations directly impact demand for products and services. However, many businesses are unable to accurately track these fluctuations due to the absence of proper analytical frameworks. As a result, they struggle to recognize trends such as peak sales periods, low-demand seasons, and shifts in customer buying behaviour.

Another critical aspect of the problem is the difficulty in managing and organizing large datasets. Sales data often consists of multiple attributes such as product categories, regions, pricing, and transaction dates. Handling such complex data manually or through basic tools becomes inefficient and error-prone. This leads to inconsistencies in data interpretation and limits the ability of organizations to derive accurate conclusions. Furthermore, businesses face challenges in understanding the contribution of different factors to overall sales performance.

It becomes difficult to determine which product categories are performing well, which regions are generating higher revenue, and which time periods contribute most significantly to sales. The absence of clear visibility into these aspects hinders strategic planning and decision-making. Another dimension of the problem is related to the lack of effective communication of data insights. Even when data is available, presenting it in a meaningful and understandable format remains a challenge. Traditional methods of reporting, such as static tables and basic charts, fail to convey complex patterns and relationships within the data. This limits the ability of stakeholders to interpret information and make informed decisions.

In addition, the dynamic nature of the market further complicates the problem. Customer preferences change rapidly, and businesses must adapt to these changes to remain competitive. Without a proper understanding of past sales trends and seasonal patterns, it becomes difficult to anticipate future demand or align business strategies accordingly. Overall, the broad problem can be summarized as the inability of organizations to effectively analyze, interpret, and understand seasonal sales data due to the complexity of datasets, lack of structured analysis, and limited visibility into key performance factors. This problem significantly affects business performance, decision-making processes, and the ability to respond to market changes.

## 1.3. Identification of Tasks

The successful completion of the Seasonal Sales Analysis project requires a systematic and well-structured approach. The entire project is divided into multiple tasks, each focusing on a specific phase of development. These tasks are carefully defined to ensure proper identification of requirements, development of the solution, and validation of results. The tasks can broadly be categorized into three major phases: **Identification Phase, Development Phase, and Testing & Validation Phase.**

### 1.3.1 Identification Phase

The identification phase is the initial stage of the project, where the problem and requirements are clearly understood. This phase plays a crucial role in defining the scope and direction of the project.

**Key Tasks:**

- **Problem Understanding:**Analyze the issue related to seasonal sales and identify key challenges such as fluctuating demand and lack of trend visibility.
- **Requirement Analysis:** Identify what the client needs, such as sales insights, seasonal

trends, and performance indicators.

- **Data Source Identification:** Select an appropriate dataset (Kaggle online sales dataset) that contains relevant attributes like product category, region, revenue, and date.
- **Defining Objectives:** Establish clear goals such as analyzing trends, identifying high-performing categories, and understanding regional performance.

### 1.3.2 Development Phase

The development phase involves building the solution using Power BI. This phase includes multiple technical tasks related to data handling, modelling, and visualization.

**Key Tasks:**

**Data Collection:** Import the dataset into Power BI for analysis.

**Data Cleaning:**

Remove duplicate records, handle missing values, and ensure data consistency using Power Query.

**Data Transformation:**

Create additional fields such as month, year, and season to enable time-based analysis.

**Data Modeling:**

Establish relationships between tables and organize the data structure for efficient querying.

**DAX Calculations:**

Create measures such as total revenue, total units sold, and growth percentage using DAX.

**Dashboard Design:**

Develop interactive visualizations including:

- KPI Cards.
- Line Charts.
- Bar Charts.
- Pie charts.

**User Interface Design:**

Ensure the dashboard is user-friendly, visually appealing, and easy to interpret.

### 1.3.3 Testing and Validation Phase

This phase ensures that the developed solution is accurate, reliable, and meets the project objectives.

**Key Tasks:**

**Data Validation:**

Verify the correctness of calculations and ensure data integrity.

**Functionality Testing:**

Ensure that filters, slicers, and interactions work correctly.

**Result Verification:**

Compare dashboard outputs with raw data to confirm accuracy.

**User Testing:**

Evaluate whether the dashboard is easy to understand and useful for decision-making.

### 1.3.4 Documentation and Reporting

Proper documentation is an essential part of the project. It ensures that the methodology, analysis, and results are clearly communicated.

**Key Tasks:**

- Preparing detailed project report
- Structuring chapters and sections
- Adding screenshots of dashboards
- Explaining visualizations and insights

### 1.3.5 Framework of the Report

The project report is organized into the following chapters:

- **Chapter 1: Introduction -** Covers problem identification, objectives, and tasks.
- **Chapter 2: Literature Review -** Covers problem identification, objectives, and tasks.
- **Chapter 3: Design Flow/ Process -** Explains methodology, system design, and implementation steps.
- **Chapter 4: Result Analysis -** Presents dashboard outputs and insights.
- **Chapter 5: Conclusion and Future Work -** Summarizes findings and suggests improvements.

## 1.4. Timeline

The successful execution of the Seasonal Sales Analysis project requires a well-defined timeline to ensure that all tasks are completed systematically within the allocated time frame. A structured timeline helps in effective planning, monitoring progress, and ensuring timely completion of each phase of the project. For this purpose, a Gantt chart is used to represent the schedule of activities in a visual and organized manner.

The project is divided into multiple phases, including data collection, data preprocessing, development, testing, and documentation. Each phase is assigned a specific duration based on its complexity and importance

**Project Timeline Description**

The project begins with the data collection phase, where the dataset is obtained from Kaggle. This phase typically requires 1–2 days to understand the dataset structure and attributes.
The next phase is data cleaning and preprocessing, which involves removing missing values, correcting inconsistencies, and transforming data into a suitable format. This phase is crucial and requires approximately 2–3 days. Following this, the data modelling and development phase is carried out. In this phase, relationships between data fields are established, and DAX measures are created.
Additionally, dashboards are designed using various visualizations such as charts and graphs. This phase takes around 4–5 days as it involves both technical and design aspects.
The testing and validation phase ensures that the dashboard is accurate and functions correctly. It includes verifying calculations, testing filters and slicers, and ensuring data consistency. This phase requires 2–3 days. Finally, the documentation phase is completed, where the project report is prepared, including all chapters, explanations, and screenshots. This phase usually takes 5–7 days as it involves detailed writing and formatting**.**

**Gantt Chart Representation :-**

*Table 1 : Gantt chart Representation*

| **Task** | **Duration** | **Week** |
|---|---|---|
| Data Collection | 1–2 days | Week 1 |
| Data Cleaning & Preprocessing | 2–3 days | Week 1 |
| Data Modelling | 2 days | Week 2 |
| Dashboard Development | 3 days | Week 2 |
| Testing & Validation | 2–3 days | Week 3 |
| Report Writing & Documentation | 5–7 days | Week 3 |

## 1.5. Organization of the Report

Give The project report titled *"Seasonal Sales Analysis using Power BI"* is systematically organized into multiple chapters to ensure a clear, logical, and comprehensive presentation of the work carried out. Each chapter focuses on a specific aspect of the project, starting from problem identification to final conclusions and future scope. This structured approach helps readers understand the methodology, analysis, and results in a sequential manner.

**Chapter 1: Introduction**

This chapter provides an overview of the project by introducing the problem and its relevance in the current business environment. It includes the identification of the client and the need for the project, along with a discussion of the contemporary issues related to seasonal sales analysis. The chapter also defines the problem statement, outlines the tasks involved in the project, presents the timeline, and explains how the report is organized. This chapter sets the foundation for the entire project.

**Chapter 2: Literature Review / Background Study**

This chapter focuses on the theoretical background and existing research related to the problem. It discusses the evolution of sales analysis techniques and highlights various tools and methods used in the past, such as spreadsheets, traditional business intelligence tools, and advanced analytics techniques. A comparative analysis of these solutions is provided, along with their advantages and limitations. The chapter concludes by linking the findings of the literature review with the current project and clearly defining the problem and objectives.

**Chapter 3: Design Flow / Process**

This chapter describes the methodology adopted to develop the solution. It includes the evaluation and selection of features required for the project, along with the constraints considered during the design process. The chapter explains the data preprocessing steps, data modelling techniques, and DAX calculations used in Power BI. It also presents the design flow, including alternative approaches and the rationale behind selecting the final design. Additionally, the implementation plan and system workflow are explained in detail.

**Chapter 4: Results Analysis and Validation**

This is one of the most important chapters of the report, as it presents the actual outcomes of the project. It includes detailed explanations of the Power BI dashboard and its various components, such as KPI indicators, monthly sales trends, seasonal comparisons, category-wise analysis, regional distribution, and payment methods. Each visualization is interpreted to derive meaningful insights. The chapter also includes validation of results through testing and verification to ensure accuracy and reliability.

**Chapter 5: Conclusion and Future Work**

This chapter summarizes the overall findings of the project and evaluates whether the objectives have been successfully achieved. It highlights the key insights obtained from the analysis and discusses the impact of the project on decision-making processes. The chapter also identifies any limitations encountered during the project. In addition, it provides suggestions for future enhancements, such as incorporating predictive analytics, real-time data processing, and advanced visualization techniques.

# CHAPTER 2

# LITERATURE REVIEW/BACKGROUND STUDY

## 2.1. Timeline of the reported problem

The problem of analyzing sales data and identifying seasonal patterns has been recognized globally for several decades. With the growth of retail and e-commerce industries, understanding customer behaviour and sales trends has become increasingly important. The evolution of this problem can be traced through different stages:

- **Early-1990s:**
  During this period, businesses relied mainly on manual record-keeping and basic spreadsheet tools for sales tracking. Seasonal trends were identified through observation rather than systematic analysis. There was limited use of data for decision-making.
- **2000–2010:**
  The introduction of database management systems and enterprise resource planning (ERP) tools allowed organizations to store and manage large volumes of sales data. However, analysis was still limited to static reports, which lacked interactivity and real-time insights.
- **2010–2020:**
  The emergence of Business Intelligence (BI) tools such as Microsoft Power BI, Tableau, and Qlik transformed the way businesses analyze data. These tools enabled interactive dashboards, allowing users to visualize trends and patterns more effectively. Seasonal sales analysis became more structured and data-driven.
- **2020–Present:**
  With advancements in data analytics, artificial intelligence, and cloud computing, organizations now use advanced tools for predictive analysis and real-time monitoring. Seasonal sales analysis has become critical for inventory management, marketing strategies, and revenue optimization.

According to industry reports, organizations that fail to analyze sales trends effectively may experience up to **20–30% loss in revenue due to poor forecasting and planning**. This highlights the importance of using modern tools to analyze seasonal sales patterns.

## 2.2. Existing solutions

Over the years, several approaches have been developed to analyze sales data and identify trends. These solutions vary in complexity, usability, and effectiveness.

**1. Spreadsheet-Based Analysis (Microsoft Excel)**

Spreadsheet tools have been widely used for analyzing sales data.

- **Applications:**
  - Data entry and storage
  - Pivot tables and charts
  - Basic calculations
- **Advantages:**
  - Easy to use
  - Widely available
  - Suitable for small datasets
- **Limitations:**
  - Not scalable for large datasets
  - Limited interactivity
  - Manual effort required

**2. Traditional Business Intelligence Tools**

Tools such as Tableau, QlikView, and SAP BusinessObjects are used for advanced data visualization.

- **Applications:**
  - Dashboard creation
  - Data visualization
  - Reporting
- **Advantages:**
  - High-quality visualizations
  - Better performance
- **Limitations:**
  - Expensive
  - Requires technical expertise

**3. Machine Learning-Based Systems**

Machine learning models are used for predictive analysis.

- **Applications:**
  - Demand forecasting
  - Customer behaviour prediction
- **Advantages:**
  - High accuracy
  - Handles large datasets
- **Limitations:**
  - Complex implementation
  - Difficult to interpret

**4. Power BI (Selected Approach)**

Power BI is a modern business intelligence tool used for interactive data visualization.

- **Applications:**
  - Dashboard creation
  - Real-time reporting
  - Data transformation
- **Advantages:**
  - User-friendly
  - Cost-effective
  - Interactive
- **Limitations:**
  - Limited advanced analytics
  - Requires DAX knowledge

## 2.3. Bibliometric analysis

Bibliometric analysis involves evaluating existing research and solutions based on key parameters such as features, effectiveness, and drawbacks.

★ **Key Features Identified**

- Data visualization
- Trend analysis
- Interactivity
- Data modelling
- Dashboard creation

★ **Effectiveness of Existing Solutions**

- Excel is effective for small datasets but lacks scalability
- BI tools provide better visualization and insights
- Machine learning offers predictive capabilities

★ **Drawbacks Identified**

- Complexity of advanced tools
- Lack of real-time insights
- High cost of some BI tools
- Limited accessibility for non-technical users

***Table 1: Comparative Analysis***

| Feature | Excel | Tableau | Power BI | ML Model |
|---|---|---|---|---|
| Ease of use | High | Medium | High | Low |
| Cost | Low | High | Medium | High |
| Interactivity | Low | High | High | Medium |
| Scalability | Low | High | High | High |

The analysis shows that while each tool has its strengths, Power BI provides the best balance between usability, cost, and functionality.

## 2.4. Review Summary

From the literature review, the following conclusions are drawn:

- Advertising spend Vs sales analysis is essential for business growth
- Traditional methods are insufficient for modern data needs
- Advanced tools improve decision-making
- Visualization plays a key role in understanding data.

**Link to Current Project**

This project builds upon the findings of the literature review by:

- Using Power BI for data visualization
- Creating an interactive dashboard
- Analyzing Advertising spend Vs sales patterns

Thus, the project aims to provide a practical solution for understanding sales trends.

## 2.5. Problem Definition

Based on the literature review, the problem is defined as:

The inability of organizations to effectively analyze advertising spend and sales data and identify patterns due to lack of proper visualization tools and structured analysis.

★ **What is to be Done**

- Analyze sales dataset
- Identify advertising trends
- Create visualizations

★ **How it is to be Done**

- Use Power BI
- Apply data preprocessing
- Create dashboards

★ **What is NOT to be Done**

- No machine learning prediction
- No real-time data integration

- No modification of dataset beyond cleaning

## 2.6. Goals/Objectives

The goals and objectives of the *Advertising spend Vs Sales Analysis using Power BI* project are defined to provide clear direction and measurable outcomes throughout the course of the project. These objectives are carefully designed to be specific, precise, tangible, and achievable, ensuring that the project remains focused and aligned with its intended purpose.

- **Primary Objectives:**

The primary objectives of this project focus on analyzing sales data and extracting meaningful insights through visualization:

- **To analyze seasonal sales trends:** Study how sales vary across different months and seasons to identify patterns and fluctuations in demand.
- **To develop an interactive Power BI dashboard:** Design and implement a user-friendly dashboard that allows users to explore data dynamically using filters and slicers.
- **To identify high-performing products and categories:** Determine which product categories and individual products contribute the most to total revenue.
- **To analyze regional sales performance:** Evaluate how sales vary across different geographical regions and identify key markets.
- **To examine customer payment behaviour:** Analyze different payment methods to understand customer preferences and transaction patterns.

**Secondary Objectives:**

These objectives support the primary goals and enhance the overall quality of the project:

- **To perform data cleaning and preprocessing:** Ensure that the dataset is accurate, consistent, and suitable for analysis.
- **To apply data modelling techniques:** Establish relationships between different data fields and create calculated measures using DAX.
- **To create meaningful and effective visualizations:** Use charts such as line graphs, bar charts, and pie charts to represent data clearly.

- **To improve data interpretation skills:** Develop the ability to interpret visual data and derive actionable insights.

**Measurable Outcomes:**

The objectives of the project are designed to be measurable and verifiable through the following outcomes:

- Successful creation of an interactive dashboard with multiple visualizations
- Accurate calculation of key performance indicators (KPIs) such as total revenue and units sold
- Identification of seasonal patterns and trends in sales data
- Clear representation of insights through charts and graphs
- Validation of results by comparing outputs with the dataset

**Characteristics of Objectives:**

The objectives of this project follow the SMART criteria:

- **Specific:** Focused on seasonal sales analysis and visualization
- **Measurable:** Based on dashboard outputs and insights generated
- **Achievable:** Can be accomplished using Power BI tools
- **Relevant:** Addresses real-world business problems
- **Time-bound:** Completed within the defined project timeline

# CHAPTER 3

# DESIGN FLOW/PROCESS

## 3.1. Evaluation & Selection of Specifications/Features

The development of an effective Advertising spend Vs Sales Analysis dashboard requires careful identification and evaluation of features that can provide meaningful insights. Based on the literature review and understanding of business intelligence systems, several features were identified and critically analyzed. The selection of features is guided by their relevance, usability, efficiency, and ability to support decision-making.

**Identified Features from Literature**

From previous studies and existing solutions, the following key features were identified:

**Data visualization using charts and graphs**

- Interactive dashboards
- Time-based trend analysis
- Category-wise and product-wise comparison
- Region-wise analysis
- Customer payment behaviour analysis
- KPI indicators for summary statistics
- Filtering and slicing of data

These features are commonly used in business intelligence tools to analyze large datasets and present insights in an understandable format.

**Critical Evaluation of Features**

**Each feature was evaluated based on its importance, advantages, and limitations.**

*Table 3: Critical Evaluation of Features*

| Feature | Importance | Advantage | Limitations |
|---|---|---|---|
| Data Visualization | High | Easy Interpetation of Data | Require Proper Design |
| Interactivity | High | User-friendly and dynamic | Can Increase Complexity |
| Trend Analysis | High | Identifies seasonal Patterns | Needs time-based data |
| Category Analysis | Medium | Product performance insights | May Cause Clutter |
| Region Analysis | High | Market performance understanding | Depends on data availability |
| Payment Analysis | Medium | Customer Behaviour Insights | Limited Impact of Core Analysis |
| KPI Indicators | High | Quick Summary of Perfommance | Limited detail |
| Filters/Slicers | High | Flexible Data Exploration | Required Structured Data |

**Analysis of Feature Effectiveness**

- **Data Visualization** is essential as it converts raw data into meaningful graphical representations, making it easier to interpret complex datasets.
- **Interactivity** enhances user experience by allowing dynamic filtering and exploration of data.
- **Trend Analysis** is crucial for identifying seasonal patterns, which is the core objective of this project. **Category and Product Analysis** help in understanding which products contribute most to revenue.
- **Region Analysis** provides insights into geographical performance, which is useful for business expansion strategies.
- **KPI Indicators** offer a quick overview of key metrics such as total revenue and units sold.

**Selected Features**

1. **Monthly Sales Trend Analysis (Line Chart)**
    - To identify variations in sales over time
2. **Seasonal Sales Comparison (Bar Chart)**
    - To compare performance across different seasons
3. **Revenue by Product Category (Bar Chart)**
    - To analyze contribution of each category
4. **Sales Distribution by Region (Pie Chart)**
    - To understand geographical performance
5. **Product-Level Revenue Analysis (Horizontal Bar Chart)**
    - To identify top-performing products
6. **Payment Method Analysis (Donut Chart)**
    - To analyze customer transaction behavior
7. **KPI Cards (Summary Indicators)**
    - Total Revenue
    - Total Units Sold
    - Unit Price
8. **Filters and Slicers**
    - Region
    - Category
    - Time (Month/Year)

**Justification of Selected Features**

The selected features are ideal for the following reasons:

- They provide **comprehensive analysis** of sales data
- They support **interactive exploration** of data
- They are **easy to understand** for both technical and non-technical users
- They align with the **objective of seasonal sales analysis**

- They ensure **efficient decision-making** through clear insights

**Rejected / Excluded Features**

Certain features were not included due to project constraints:

- Machine learning-based prediction models
- Real-time data analytics
- Complex statistical modelling

These were excluded to maintain simplicity and focus on visualization.

## 3.2. Design Constraints

The design and development of the Advertising spend vs Sales Analysis dashboard are influenced by various constraints that ensure the system is practical, ethical, and aligned with professional standards. These constraints help define the scope of the project and guide the decision-making process during implementation.

**Standards Considered in the Design:**

The following standards and considerations were taken into account while designing the system:

**1. Economic Constraints**

Economic constraints refer to the financial limitations associated with the project.

- The project is developed using the free/student version of Microsoft Power BI
- No additional paid tools or software were used
- The solution is designed to be cost-effective and accessible

Impact:

- Limits access to advanced premium features
- Ensures the project is feasible for academic and small-scale use

**2. Environmental Constraints**

Environmental considerations are important for sustainable system design.

- The dashboard is completely digital, reducing the need for paper-based reports
- Promotes eco-friendly practices by minimizing resource usage
- No physical infrastructure or materials are required

Impact:

- Supports sustainability
- Reduces environmental footprint

**3. Health and Safety Constraints**

Although the project is not directly related to physical health systems, certain considerations are still important.

- Ensures that data handling does not create any risk
- Avoids misuse of sensitive information
- Maintains safe digital practices

Impact:

- Protects data integrity
- Prevents misuse of information

**4. Ethical Constraints**

Ethical considerations are critical in any data-related project.

- Data is used without manipulation or falsification
- Insights are presented honestly without bias
- No misleading visualizations are created

Impact:

- Maintains credibility of analysis
- Ensures trustworthiness of results

**5. Social Constraints**

The system must be accessible and understandable to a wide range of users.

- Dashboard is designed for both technical and non-technical users
- Uses simple and clear visualizations
- Avoids complex terminology

Impact:

- Improves usability

- Enhances user experience

**6. Professional Constraints**

The project must adhere to academic and professional standards.

- Proper report structure is followed
- Use of formal language and documentation
- Inclusion of all required sections (introduction, methodology, results, etc.)

Impact:

- Ensures academic quality
- Meets evaluation criteria

**7. Technical Constraints**

Technical limitations affect system design and implementation.

- Dataset is static and limited in size
- No real-time data integration
- Limited advanced analytics capabilities

Impact:

- Restricts complexity of analysis
- Focuses on visualization

**8. Cost Constraints**

Cost plays a major role in project feasibility.

- Zero-cost implementation using free tools
- No hardware or software investment required

Impact:

- Makes the project accessible
- Suitable for students and small organizations

**9. Political and Regulatory Constraints**

Although the project does not directly involve government policies, certain general guidelines are considered:

- No violation of data usage policies
- Dataset is publicly available and legally usable
- No sensitive or confidential data is used

Impact:

- Ensures legal compliance
- Avoids misuse of data

**Overall Impact of Constraints**

The above constraints collectively influence the design of the system by:

- Defining the scope and limitations of the project
- Ensuring ethical and responsible data usage
- Promoting simplicity and usability
- Maintaining cost-effectiveness
- Aligning with academic and professional standards

Rather than restricting the project, these constraints help in creating a balanced, practical, and efficient solution that meets the objectives of seasonal sales analysis.

## 3.3. Analysis of Features and finalization subject to constraints

After identifying and evaluating various features in the previous section, it is essential to refine and finalize these features based on the design constraints. The constraints such as technical limitations, economic feasibility, simplicity, and usability play a crucial role in determining which features should be included, modified, or excluded. This process ensures that the final system is practical, efficient, and aligned with the objectives of the project.

**Need for Feature Finalization**

Although several advanced features were identified during the literature review, not all of them are suitable for implementation within the scope of this project. The goal is to design a system that is:

- Easy to understand
- Cost-effective
- Technically feasible

- Focused on visualization

Therefore, features were carefully analyzed and categorized into three groups: Removed Features, Modified Features, and Added Features.

**1. Removed Features**

Certain features were excluded due to constraints such as complexity, lack of data, or irrelevance to project objectives.

**a) Machine Learning-Based Prediction**

- Not included due to complexity and scope limitations
- Requires advanced programming and additional tools
- Not aligned with visualization-focused objective

**b) Real-Time Data Integration**

- Dataset used is static (Kaggle dataset)
- No API or live data source available
- Real-time systems require additional infrastructure

**c) Complex Statistical Modelling**

- Techniques such as regression and hypothesis testing excluded
- Difficult to interpret for non-technical users
- Not necessary for basic trend analysis

**2. Modified Features**

Some features were adjusted to fit within the constraints while still providing meaningful insights.

**a) Trend Analysis**

- Modified to use available time-based data (Month, Ad spend)
- Instead of advanced forecasting, simple trend visualization is used

**b) Data Visualization**

- Simplified charts used instead of complex visuals
- Focus on clarity and readability

**c) Category and Product Analysis**

- Limited to available dataset fields

- Focus on key categories rather than excessive breakdown

**3. Added Features**

To enhance the effectiveness of the dashboard, some additional features were introduced.

**a) KPI Cards (Key Performance Indicators)**

- Total Revenue
- Total Units Sold
- Unit Price

Purpose:

Provide quick summary of business performance

**b) Seasonal Comparison**

- Comparison of sales across different seasons

Purpose:

Identify peak and low-demand periods

**c) Region-Wise Analysis**

- Sales distribution by region

Purpose:

Understand geographical performance

**d) Payment Method Analysis**

- Analysis of customer payment preferences

Purpose:

Identify popular payment methods

**e) Filters and Slicers**

- Enable dynamic data exploration

Purpose:

Improve interactivity and user experience

**Final Feature Set**

After analysis and refinement, the final set of features includes:

- Monthly Sales Trend

- Seasonal Sales Comparison
- Revenue by Category
- Region-Wise Sales Analysis
- Product-Level Analysis
- Payment Method Analysis
- KPI Indicators
- Filters and Slicers

## 3.4. Design Flow

The design flow represents the systematic process followed to develop the Advertising spend Vs Sales Analysis solution. It outlines the sequence of steps required to transform raw data into meaningful insights. To ensure an optimal solution, two alternative design approaches were considered and evaluated before selecting the final implementation.

**Need for Multiple Design Approaches**

Considering multiple design approaches helps in:

- Comparing different methods of implementation
- Identifying the most efficient and effective solution
- Ensuring that the selected design meets project requirements

In this project, two design approaches were evaluated:

**1.Traditional Data Analysis Approach**

**2.Power BI Dashboard Approach (Selected)**

**Design 1: Traditional Data Analysis Approach**

**Process Flow**

1. Data Collection (Kaggle dataset)
2. Data Cleaning using Excel
3. Manual Data Analysis
4. Chart Creation using Excel
5. Static Report Generation

**Explanation of Steps**

- Data Collection: Dataset is imported into Excel
- Data Cleaning: Missing values are handled manually

- Data Analysis: Calculations are performed using formulas
- Chart Creation: Graphs are created using Excel tools
- Report Generation: Results are presented in static form.

**-Advantages**

- Simple and easy to implement
- Requires basic tools
- Suitable for small datasets

**-Disadvantages**

- Static and non-interactive
- Time-consuming manual work
- Difficult to analyze large datasets
- Limited visualization capabilities

**-Design 2: Power BI Dashboard Approach**

**Process Flow**

1. Data Collection (Kaggle dataset)
2. Data Import into Power BI
3. Data Cleaning using Power Query
4. Data Modelling
5. DAX Calculations
6. Visualization Creation
7. Interactive Dashboard Development
8. Analysis and Insights

**-Explanation of Steps**

- **Data Import:** Dataset is loaded into Power BI
- **Data Cleaning:** Performed using Power Query Editor
- **Data Modelling:** Relationships between fields are created
- **DAX Calculations:** Measures like revenue and units sold are defined
- **Visualization:** Charts such as line, bar, and pie charts are created
- **Dashboard Development:** Interactive dashboard is designed
- **Analysis:** Insights are derived from visualization

**-Advantages**

- Interactive and dynamic dashboard
- Easy to analyze large datasets
- Advanced visualization capabilities
- Supports filtering and slicing
- Faster and more efficient analysis

**-Disadvantages**

- Requires knowledge of Power BI
- Initial setup is slightly complex

**Comparison of Design Approaches**

*Table 4: Comparison of Design Approaches*

| Criteria | Traditional Method | Power BI Approach |
|---|---|---|
| Interactivity | No | Yes |
| Visualization Quality | Medium | High |
| Ease of Analysis | Low | High |
| Efficiency | Low | High |
| Scalability | Low | High |
| User Experience | Basic | Advanced |

**Selected Design**

Based on the comparison, the Power BI Dashboard Approach was selected as the final design for this project.

- Reason for Selection
- Provides interactive and dynamic analysis
- Enhances data visualization and clarity
- Supports real-time filtering and exploration
- Aligns with modern business intelligence practices
- Improves decision-making efficiency

**Final Design Flow**

The final design flow of the system is as follows:

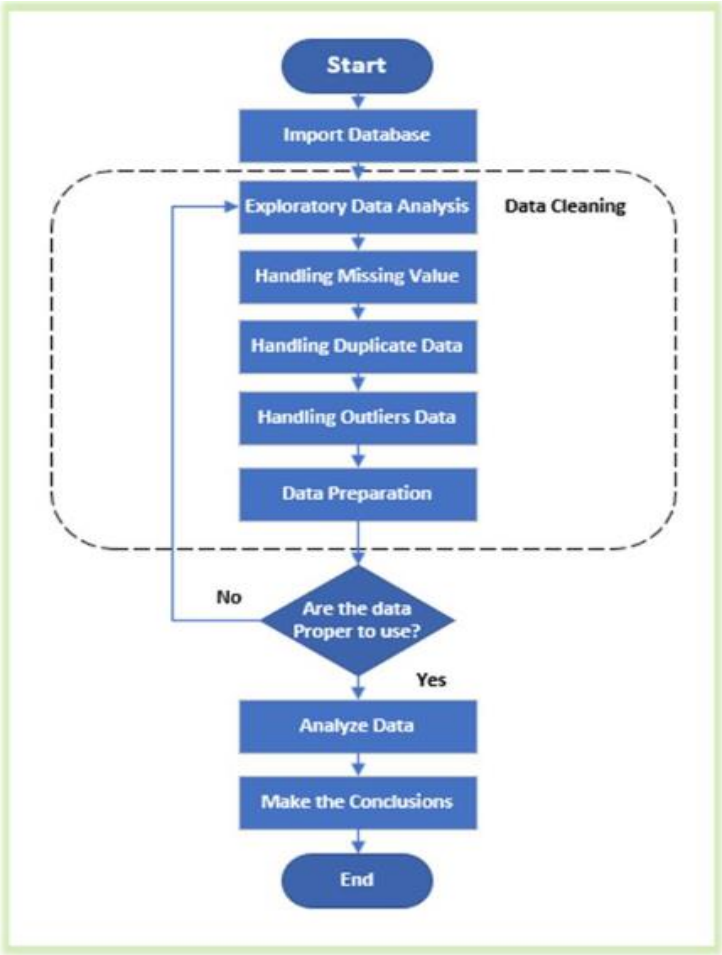



*Figure 1: Design Flow*

## 3.5. Implementation plan/methodology

The implementation plan of the Seasonal Sales Analysis project outlines the step-by-step procedure followed to transform raw data into meaningful insights using Microsoft Power BI. This methodology ensures a structured approach to data processing, visualization, and analysis.

The implementation consists of multiple stages, including data collection, preprocessing, modeling, visualization, and validation.

**1.Step-by-Step Implementation Process**

**Step 1: Data Collection**

The dataset is obtained from Kaggle

It contains attributes such as:

- Product Category
- Product Name
- Region
- Revenue
- Unit Price
- Total Units Sold
- Payment Method

**Purpose:**

To gather relevant data required for analysis.

**Step 2: Data Preprocessing**

This step ensures that the dataset is clean and ready for analysis.

**Tasks performed:**

- Removal of null values
- Elimination of duplicate records
- Correction of data types
- Formatting of columns

**Step 3: Data Transformation**

Additional fields are created to support analysis:

- Month
- Year
- Season

**Purpose:**

To enable time-based and seasonal analysis.

**Step 4: Data Modelling**

- Relationships between data fields are defined
- Measures are created using DAX:
    - Total Revenue
    - Total Units Sold

**Purpose:**

To structure data for efficient querying and analysis.

**Step 5: Visualization Creation**

Various charts are created:

- Line Chart (Monthly Trend)
- Bar Chart (Category Analysis)
- Pie Chart (Region Analysis)
- Donut Chart (Payment Analysis)

**Step 6: Dashboard Development**

- All visualizations are combined into a single dashboard
- Layout is designed for clarity and usability
- KPI cards are added

**Step 7: Interactivity Implementation**

- Filters and slicers are added
- Users can explore data dynamically

**Step 8: Testing and Validation**

- Verify correctness of calculations
- Check functionality of filters
- Ensure consistency of data

**Step 9: Analysis and Insight Generation**

- Interpret graphs
- Identify trends and patterns
- Generate business insights

## 2.Flowchart

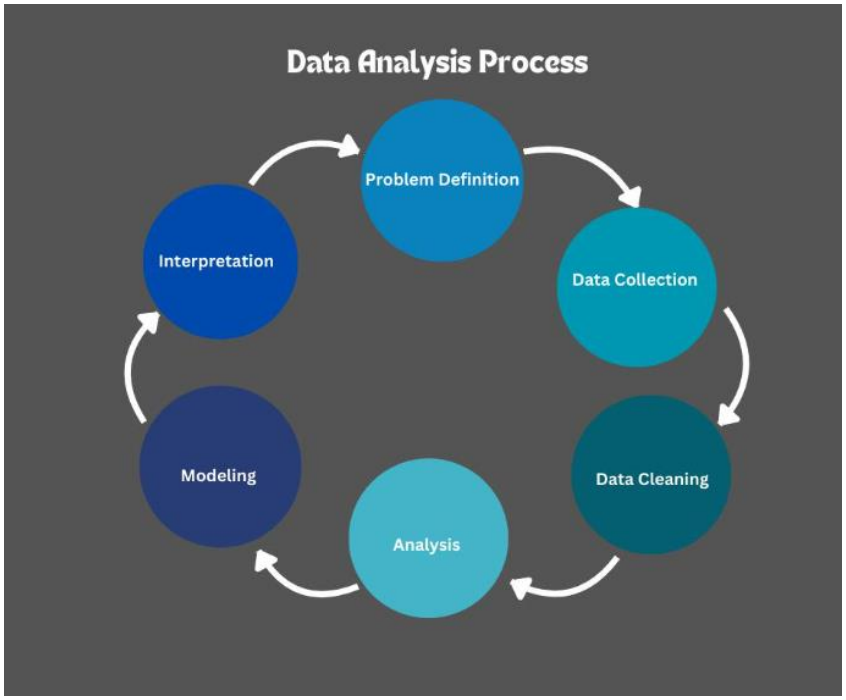



*Figure 2: Data Analysis Process*

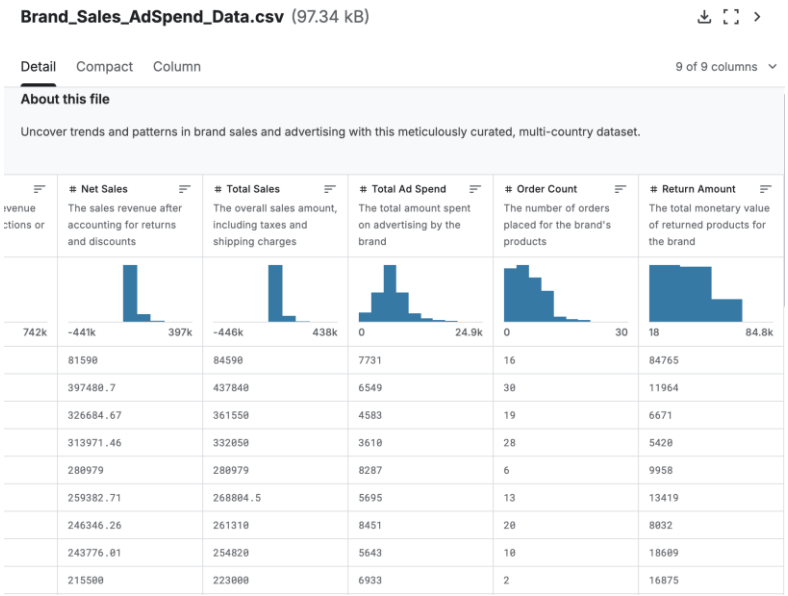
## 3.Algorithm

Algorithm for Seasonal Sales Analysis

- Start
- Load dataset from Kaggle
- Clean dataset (remove null and duplicates)
- Transform data (create Month, Year, Season)
- Create measures (Revenue, Units Sold)
- Generate visualizations
- Build dashboard
- Apply filters and slicers
- Validate results
- Analyze data
- End

## 3.6. Data Source

Link of the dataset used in the project dashboard -



**Brand_Sales_AdSpend_Data.csv** (97.34 kB)

Detail Compact Column 9 of 9 columns

**About this file**

Uncover trends and patterns in brand sales and advertising with this meticulously curated, multi-country dataset.

| | # Net Sales<br>The sales revenue after accounting for returns and discounts | # Total Sales<br>The overall sales amount, including taxes and shipping charges | # Total Ad Spend<br>The total amount spent on advertising by the brand | # Order Count<br>The number of orders placed for the brand's products | # Return Amount<br>The total monetary value of returned products for the brand |
|---|---|---|---|---|---|
| venue ctions or | -441k 397k | -446k 438k | 0 24.9k | 0 30 | 18 84.8k |
| 742k | 81590 | 84590 | 7731 | 16 | 84765 |
| | 397480.7 | 437840 | 6549 | 30 | 11964 |
| | 326684.67 | 361550 | 4583 | 19 | 6671 |
| | 313971.46 | 332050 | 3610 | 28 | 5420 |
| | 280979 | 280979 | 8287 | 6 | 9958 |
| | 259382.71 | 268804.5 | 5695 | 13 | 13419 |
| | 246346.26 | 261310 | 8451 | 20 | 8032 |
| | 243776.01 | 254820 | 5643 | 10 | 18609 |
| | 215500 | 223000 | 6933 | 2 | 16875 |

https://www.kaggle.com/datasets/shreyanshverma27/brands-sales-and-ad-spend-data

# CHAPTER 4

# RESULTS ANALYSIS AND VALIDATION

## 4.1. Implementation of solution

### Page 1 – Overview Dashboard

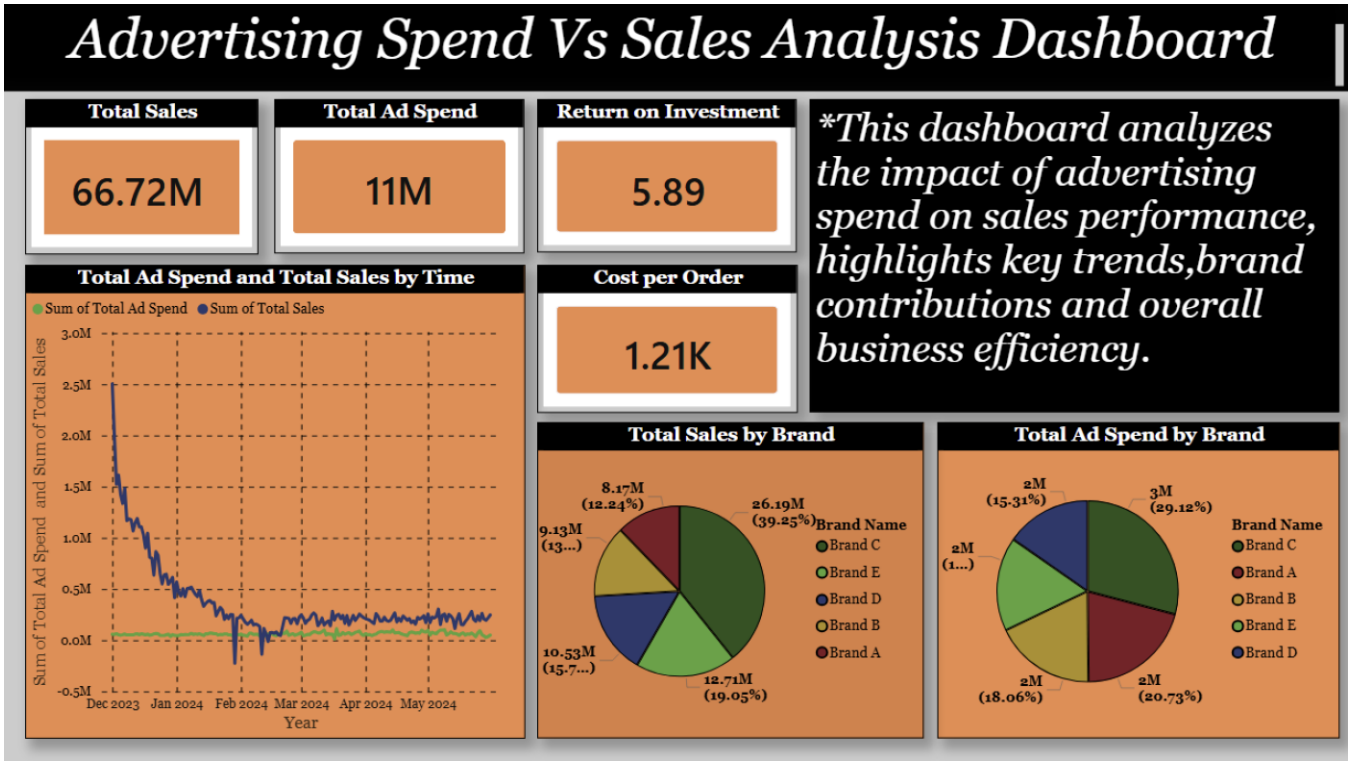



*Figure 3: Overview Dashboard*

This dashboard presents a comprehensive analysis of the relationship between advertising spend and sales performance over time. It highlights key business metrics such as total sales, total advertising expenditure, return on investment (ROI), and cost per order, providing a clear overview of overall business efficiency. By comparing advertising spend with generated sales, the dashboard helps in understanding how effectively marketing investments are being converted into revenue.

The time-based trend visualization shows fluctuations in both sales and ad spend, enabling the identification of patterns, seasonal variations, and periods of high or low performance. Additionally, the brand-wise distribution of sales and advertising spend offers deeper insights into which brands contribute the most to revenue and which require optimization in marketing strategies.

Overall, this dashboard serves as a decision-support tool for businesses, helping stakeholders evaluate marketing effectiveness, optimize budget allocation, and improve profitability through data-driven insights.

- ## Details of visualization used in Overview Dashboard

**Card – Total Sales**

The card visualization is used to display the Total Sales, which represents the overall revenue generated by the business. It provides a quick and clear snapshot of financial performance.

**Card – Total Ad Spend**

It helps in understanding how much budget has been allocated to promotions and is useful for comparing spending against revenue to evaluate marketing effectiveness.

**Card – Return on Investment (ROI)**

The ROI card highlights how efficiently advertising spend is converted into revenue. A higher ROI reflects better performance and effective utilization of marketing resources, making it a critical metric for strategic decision-making.

**Card – Cost per Order**

This card represents the average cost incurred per order, providing insight into operational efficiency. It helps businesses understand customer acquisition costs and evaluate whether processes are cost-effective.

**Line Chart – Ad Spend vs Sales Over Time**

The line chart is used to visualize trends in advertising spend and sales over time. It helps identify patterns, seasonal fluctuations, and the relationship between marketing efforts and revenue generation, supporting better forecasting and planning.

**Donut Chart – Total Sales by Brand**

The donut chart displays the proportion of total sales contributed by each brand. It enables easy comparison and highlights top-performing brands, helping businesses focus on strengthening successful

products and improving weaker ones.

**Donut Chart – Total Ad Spend by Brand**

This donut chart shows how advertising budget is distributed across different brands. It helps analyze whether spending aligns with performance, allowing for more effective allocation of marketing resources.

Page 2 – Advertising Analysis

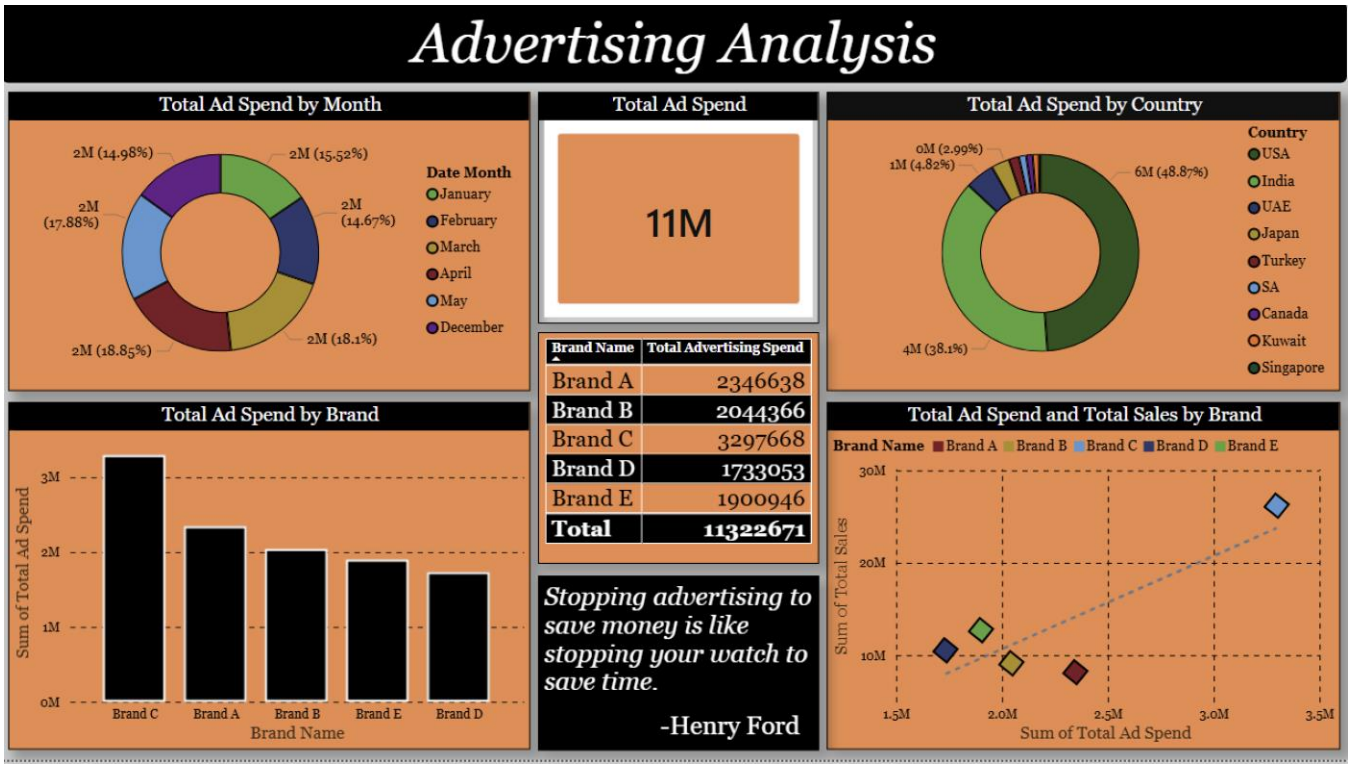



*Figure 5: Advertising Analysis Dashboard*

This dashboard provides a detailed analysis of advertising expenditure across different dimensions such as time, geography, and brand performance. It presents the total advertising spend and breaks it down month-wise, helping to identify spending patterns and seasonal trends. The monthly distribution highlights how marketing budgets are allocated throughout the year and indicates periods of higher promotional activity.

The country-wise analysis offers insights into regional investment strategies, showing which markets receive the highest share of advertising spend. This helps in understanding geographic priorities and identifying potential areas for expansion or optimization. Additionally, the brand-wise breakdown of advertising spend enables comparison among different brands, highlighting which brands are receiving more focus in terms of marketing investment.

The relationship between total ad spend and total sales by brand is visualized to evaluate the effectiveness of advertising efforts. This comparison helps in identifying high-performing brands that generate strong sales relative to their advertising costs, as well as brands that may require improved marketing strategies. Overall, this dashboard supports better decision-making by providing a clear

understanding of how advertising budgets are distributed and how effectively they contribute to business growth.

- ## Details of visualization used in Advertising Analysis

**Card – Total Ad Spend**

This card provides a summary of total advertising expenditure. It offers a quick overview of marketing investment and serves as a reference point for deeper analysis across different dimensions.

**Donut Chart – Ad Spend by Month**

This visualization shows the monthly distribution of advertising spend. It helps identify seasonal trends and variations in marketing activity, allowing businesses to plan campaigns more strategically.

**Bar Chart – Ad Spend by Brand**

The bar chart compares advertising expenditure across different brands. It clearly shows which brands receive higher investment, helping evaluate whether resources are allocated effectively.

**Table – Brand-wise Ad Spend**

The table provides detailed numerical values for advertising spend across brands. It ensures accuracy and allows for precise comparisons, making it useful for detailed reporting and analysis.

**Scatter Plot – Ad Spend vs Sales**

The scatter plot shows the relationship between advertising spend and sales for different brands. It helps identify correlation patterns and evaluate which brands are generating better returns on their marketing investments

Page 3 – Sales Analysis

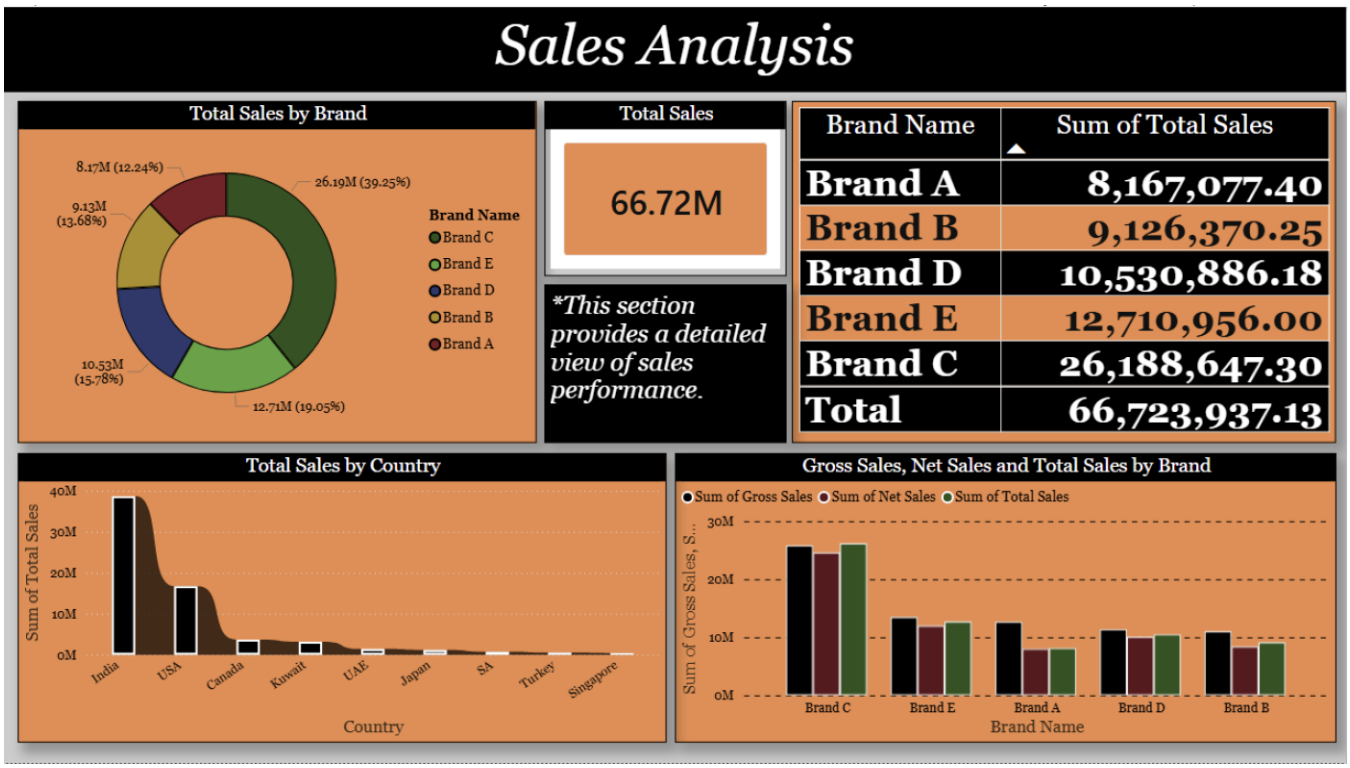



*Figure 6: Sales Analysis Dashboard*

This dashboard provides a comprehensive overview of sales performance across different brands and regions. It highlights the total sales achieved and presents a brand-wise distribution, allowing for easy comparison of each brand's contribution to overall revenue. The visualization clearly shows which brands are leading in sales and which ones have scope for improvement, helping in strategic decision-making.

The country-wise sales analysis offers insights into geographic performance, identifying key markets that drive the majority of revenue. It helps businesses understand regional demand patterns and prioritize markets for future growth and expansion. Additionally, the detailed sales table provides precise figures for each brand, ensuring transparency and accuracy in performance evaluation.

The comparison of gross sales, net sales, and total sales by brand further enhances the analysis by showing the impact of deductions such as discounts and returns. This enables a deeper understanding of actual revenue generation and profitability. Overall, this dashboard acts as an effective tool for monitoring sales performance, identifying growth opportunities, and supporting data-driven business strategies.

- ## Details of visualization used in Sales Analysis

**Card – Total Sales**

This card displays the total revenue generated by the business. It provides an immediate overview of overall sales performance and serves as a key indicator of business success.

**Donut Chart – Sales by Brand**

The donut chart illustrates the contribution of each brand to total sales. It helps compare brand performance and identify leading as well as underperforming products.

**Table – Brand-wise Sales Data**

The table presents detailed sales figures for each brand. It allows for accurate comparison and supports deeper analysis of performance metrics.

**Bar Chart – Sales by Country**

The bar chart compares sales across different countries, highlighting top-performing regions. It helps businesses understand geographic demand and prioritize markets for growth.

**Clustered Bar Chart – Gross, Net, and Total Sales**

This visualization compares multiple sales metrics side by side. It helps in understanding the impact of deductions such as discounts and returns, providing a clearer picture of actual revenue

# Page 4 – Country based Analysis

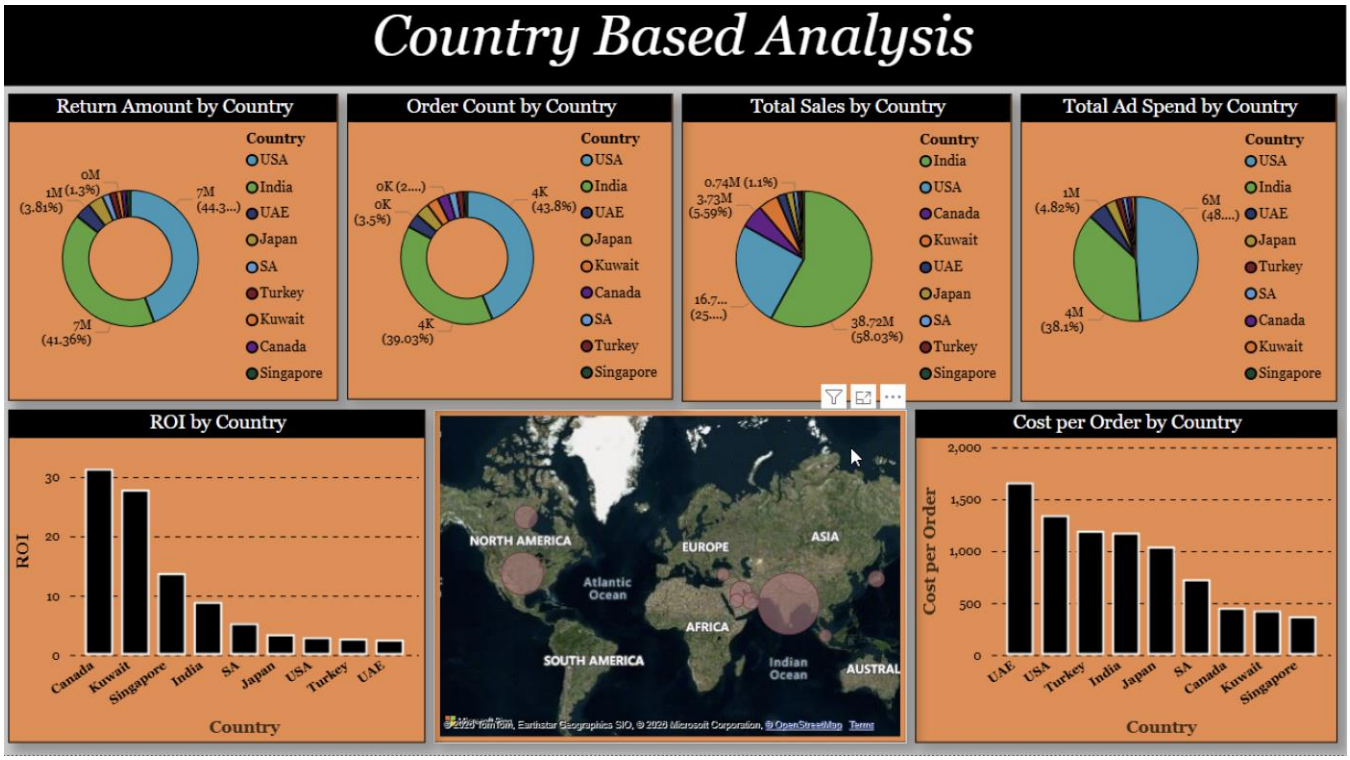



*Figure 7: Country-based Analysis*

This dashboard provides a detailed country-wise analysis of key business metrics, including total sales, advertising spend, return amount, order count, return on investment (ROI), and cost per order. It helps in understanding how different regions contribute to overall business performance and highlights variations in customer behavior across countries.

The visualizations show that certain countries contribute significantly to total sales and advertising spend, indicating strong market presence and higher business activity in those regions. The return amount and order count analysis further provide insights into customer engagement and product performance, helping identify markets with high demand as well as those with higher return rates.

The ROI comparison across countries is particularly useful in evaluating the effectiveness of advertising strategies in different regions. Countries with higher ROI indicate more efficient utilization of marketing budgets, while lower ROI suggests the need for optimization. Additionally, the cost per order metric helps in assessing operational efficiency and profitability across markets.

- ## Details of visualization used in Country based Analysis

**Donut Chart – Return Amount by Country**

This donut chart shows the proportion of return amounts across different countries. It helps identify regions with higher return rates, which may indicate issues related to product quality or customer satisfaction.

**Donut Chart – Order Count by Country**

This visualization displays the distribution of order counts across countries. It helps understand customer demand and market activity in different regions.

**Donut Chart – Total Sales by Country**

The donut chart highlights the contribution of each country to total sales. It allows easy identification of major markets and regions driving revenue.

**Donut Chart – Total Ad Spend by Country**

This chart shows how advertising budgets are distributed across countries. It helps analyze regional marketing focus and supports better investment decisions.

**Bar Chart – Cost per Order by Country**

This visualization compares the average cost per order across regions. It helps evaluate operational efficiency and identify areas where costs can be reduced.

**Map – Geographic Visualization**

The map provides a geographical representation of data across different countries. It enhances understanding by visually highlighting regional performance and making it easier to identify key markets.

Page 5 – Orders Analysis

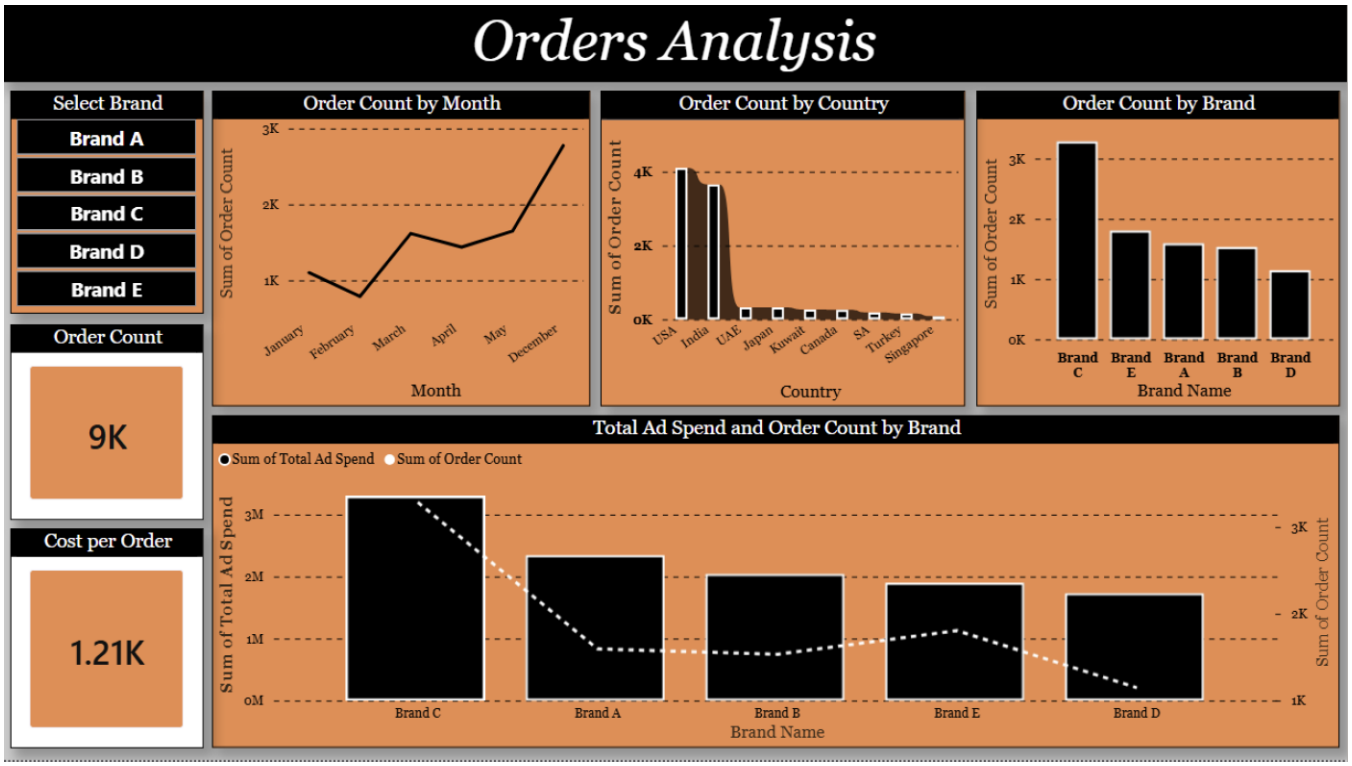



*Figure 8: Orders Analysis Dashboard*

The **Orders Analysis Dashboard** serves as a centralized intelligence hub designed to evaluate the multi-faceted performance of five distinct brands across global markets. The visual architecture is structured to move from high-level metrics to granular efficiency analysis, allowing stakeholders to immediately identify that the organization has processed a total of **9,000 orders** with a **Cost per Order of 1.21K**. By placing these KPIs prominently on the left, the dashboard establishes a baseline for success, emphasizing that while volume is high, the cost of acquisition remains a primary factor for the business to monitor. The dark-themed interface, accented with copper tones, ensures that these critical figures remain legible and command the viewer's attention before they dive into the more complex comparative data.

The temporal and geographical sections of the report highlight significant shifts in consumer behavior and market dominance. The **Order Count by Month** line graph reveals a volatile first quarter, characterized by a sharp decline in February followed by a consistent recovery. The most striking takeaway is the exponential growth observed in December, which suggests a highly successful year-end strategy or a strong seasonal demand for the products. Geographically, the data reveals a heavy reliance

on the **USA and India**, which dwarf all other regions in terms of order volume. This suggests that while the brands have a global footprint reaching from Canada to Singapore, the primary revenue drivers are concentrated in these two powerhouse markets, presenting a clear case for continued localized marketing investment in those specific regions.

- ## Details of visualization used in Orders Analysis

**Slicer – Select Brand**

The slicer is used to filter data based on selected brands. It allows users to interactively analyze order-related metrics for a specific brand, making the dashboard more dynamic and user-friendly. This helps in focusing on individual brand performance without affecting the overall dataset.

**Card – Order Count**

This card displays the total number of orders (9K), providing a quick overview of customer demand. It serves as a key indicator of business activity and helps in understanding the volume of transactions.

**Card – Cost per Order**

The card represents the average cost incurred per order. It helps evaluate operational efficiency and determine how much the business spends to fulfill each order, which is critical for profitability analysis.

**Line Chart – Order Count by Month**

The line chart shows how the number of orders changes over time. It helps identify trends, seasonal variations, and periods of high or low demand, supporting better planning and forecasting.

**Bar Chart – Order Count by Country**

This visualization compares the number of orders across different countries. It helps identify regions with the highest demand and supports decisions related to market expansion and resource allocation.

**Bar Chart – Order Count by Brand**

The bar chart presents a comparison of order counts across different brands. It highlights which brands are most popular among customers and which ones may require improvement in marketing or product strategy.

**Combo Chart (Bar + Line) – Total Ad Spend and Order Count by Brand**

This combined visualization displays both advertising spend (bars) and order count (line) for each brand. It helps analyze the relationship between marketing investment and customer demand, allowing businesses to evaluate the effectiveness of advertising strategies.

Page 6 – Overall Profitability

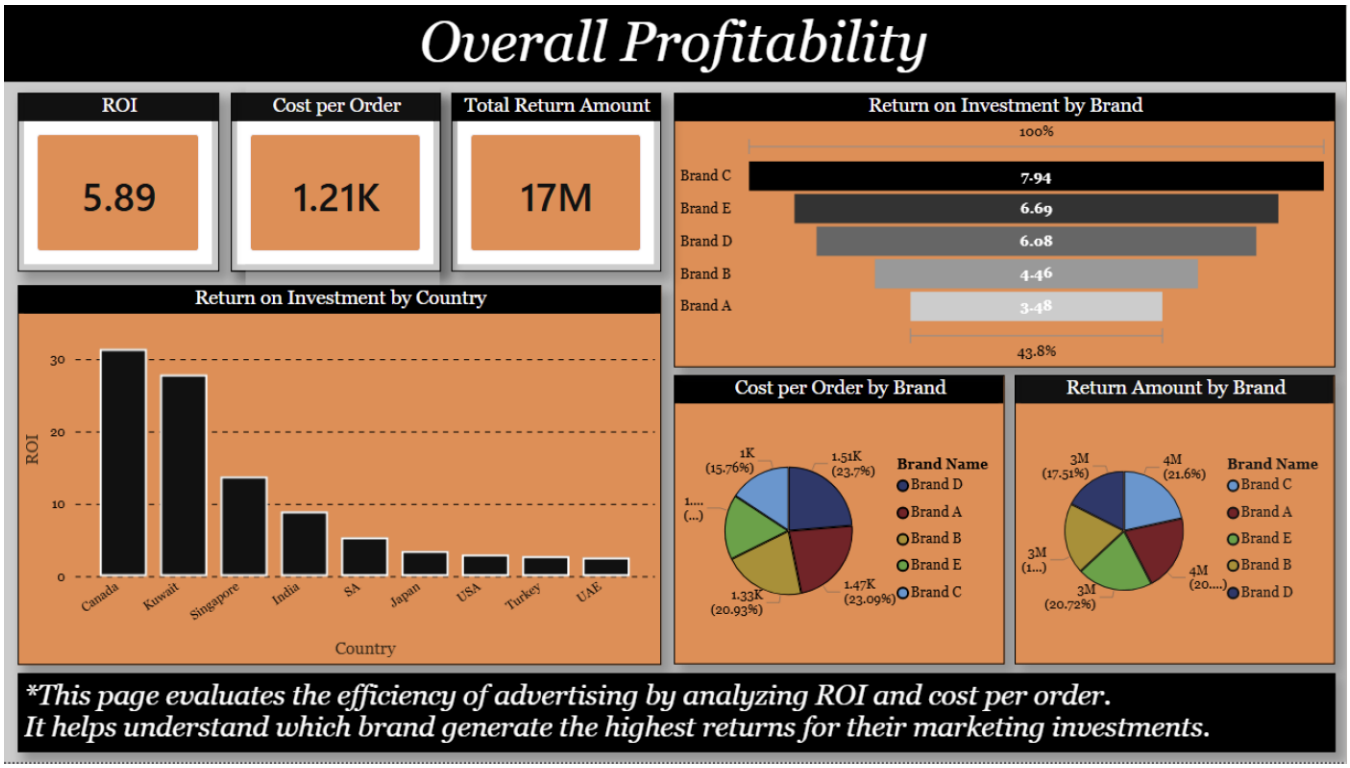



*Figure 9: Overall Profitability Dashboard*

The **Overall Profitability Dashboard** focuses on the financial efficiency of marketing efforts, shifting the perspective from pure volume to the quality of returns. At the top level, the report highlights three critical high-level metrics: a healthy **Return on Investment (ROI) of 5.89**, a consistent **Cost per Order of 1.21K**, and a substantial **Total Return Amount of 17M**. These figures provide a clear narrative of the business's current state: for every dollar spent, the company is generating nearly six times the return, resulting in a multi-million dollar revenue stream. By keeping the Cost per Order metric visible alongside the total returns, the dashboard ensures that profitability is viewed through the lens of operational efficiency rather than just gross revenue.

The **Return on Investment by Brand** chart serves as a primary benchmark for performance across the portfolio, utilizing a funnel-style visualization to rank the brands. **Brand C** is the clear leader with an impressive ROI of 7.94, followed by Brands E and D, which both maintain solid returns above 6.0. On the lower end of the spectrum, **Brand A** shows a significantly lower ROI of 3.48. When cross-referenced with the **Cost per Order by Brand** pie chart, a deeper story emerges: Brand A has a relatively high acquisition cost (1.51K), which directly impacts its lower ROI. Conversely, the **Return**

**Amount by Brand** chart indicates that while Brand C generates the highest total returns (4M or 21.6%), the other brands contribute relatively evenly to the total 17M

- ## Details of visualization used in Overall Profitability dashboard

**Card – ROI**

This card shows the overall Return on Investment, indicating how efficiently advertising spend is converted into profit. It is a key metric for evaluating business performance and marketing effectiveness.

**Card – Cost per Order**

The card displays the average cost per order, helping assess operational efficiency. Lower costs indicate better resource utilization and improved profitability.

**Card – Total Return Amount**

This card represents the total value of returned products. It helps identify the financial impact of returns and highlights potential issues related to product quality or customer satisfaction.

**Bar Chart – ROI by Country**

The bar chart compares ROI across different countries. It helps identify which regions generate higher returns from advertising investments and which require improvement.

**Bar Chart – ROI by Brand**

This visualization compares ROI across brands, highlighting the most profitable brands. It helps businesses focus on high-performing products and optimize strategies for lower-performing ones.

**Donut Chart – Cost per Order by Brand**

The donut chart shows the distribution of cost per order across brands. It helps identify which brands are more cost-efficient and which ones incur higher operational costs.

**Donut Chart – Return Amount by Brand**

This visualization displays the proportion of return amounts for each brand. It helps identify brands with higher return rates, enabling businesses to take corrective actions and i quality or customer satisfaction.

# CHAPTER 5

# CONCLUSION AND FUTURE WORK

## 5.1. Conclusion

The **Advertising and Sales Performance Dashboard** provides a multi-dimensional synthesis of market efficiency, brand health, and global scalability. By integrating financial metrics with temporal and geographic data, the report offers a definitive roadmap for optimizing future marketing investments and resource allocation.

### *Strategic Financial Efficiency*

The organization maintains a strong foundational performance with an overall **Return on Investment (ROI) of 5.89** and a stable **Cost per Order (CPO) of 1.21K**. With a **Total Return Amount of 17M** and aggregate **Total Sales of 66.72M**, the business demonstrates a high-yield model where advertising spend is effectively converted into significant top-line revenue. The data confirms that while the business invests heavily, the returns are nearly six times the initial outlay, indicating a highly profitable operational cycle.

### *Brand Performance Hierarchy*

A detailed brand-level analysis reveals a clear performance leader in **Brand C**, which dominates the portfolio across almost every metric:

- **Market Dominance:** Brand C contributes **39.25%** of total sales, amounting to **26.19M**.
- **Investment Efficiency:** It achieves the highest **ROI of 7.94**, significantly outpacing the portfolio average.
- **Scale:** It accounts for the largest share of advertising spend at **29.12% (3M)**, proving that it can absorb high investment while maintaining peak efficiency.

In contrast, **Brand A** represents a strategic challenge. Despite being the second-highest in ad spend, it

delivers the lowest ROI at **3.48** and the lowest sales contribution at **12.24%**. This disparity suggests a critical need to audit Brand A's creative strategy or audience targeting, as its cost of acquisition (**1.51K CPO**) is the highest in the portfolio.

## *Global Market Dynamics*

The geographical data presents a fascinating contrast between **volume** and **profitability**:

- **Volume Drivers: India** and the **USA** are the primary engines for scale, with India leading in total sales and the USA accounting for nearly **48.87%** of the advertising spend.
- **Profitability Gems:** While India and the USA drive volume, **Canada** and **Kuwait** are the most efficient markets. Canada, in particular, delivers a staggering ROI that dwarfs the larger markets, followed closely by Kuwait and Singapore.

This suggests a "dual-track" strategy: maintain aggressive presence in India and the USA for market share, but shift incremental budgets toward Canada and Singapore to maximize bottom-line profit margins.

The dashboard ultimately proves that the organization is in a robust growth phase, characterized by a successful end-of-year surge in December. However, to move from a 5.89 ROI to higher tiers of efficiency, the following actions are recommended:

1. **Optimization:** Reallocate underperforming ad spend from **Brand A** toward the high-efficiency model of **Brand C**.
2. **Market Expansion:** Increase penetration in **high-ROI niche markets** like Canada and Kuwait, where the return per dollar spent is significantly higher than in traditional volume centers.
3. **Efficiency Control:** Address the high **Cost per Order** in the **UAE**, which currently sees the highest acquisition costs despite moderate sales volume.

By leveraging these insights, the business can refine its "Advertising Analysis" to ensure that, as Henry Ford suggested, they are not "stopping the watch to save time," but rather winding the watch more efficiently for maximum performance.

## I. Comprehensive Performance Architecture

The dashboard's architecture is designed to bridge the gap between raw marketing expenditure and final revenue realization. With a **Total Return Amount of 17M** and **Total Sales reaching 66.72M**, the business maintains a robust financial profile. The **Return on Investment (ROI) of 5.89** serves as the primary health indicator, signaling that for every unit of currency invested in advertising, the organization recoup nearly six times that amount in value. This efficiency is mirrored by a stabilized **Cost per Order (CPO) of 1.21K**. These high-level KPIs establish that the current marketing mix is not only generating volume but is doing so with a sustainable margin that supports long-term scalability across all five brand entities.

## II. Granular Brand Portfolio Analysis

A critical segment of the report must address the divergence in brand performance, as the dashboard highlights significant variances in how each brand utilizes its allocated budget:

- **Brand C (The Market Leader):** This brand is the cornerstone of the organization's success, contributing **39.25% of total sales** (approx. 26.19M). It maintains the highest **ROI at 7.94** while absorbing the largest advertising spend of **3M (29.12%)**. This indicates a rare "high-spend, high-efficiency" profile that should be the blueprint for other brands.

- **Brand E (The Efficient Follower):** Ranking second in sales contribution at **19.05% (12.71M)** , Brand E demonstrates a strong **ROI of 6.69**. Its performance suggests a mature brand that effectively converts moderate spend into consistent market share.

- **Brand A (Strategic Optimization Needed):** Brand A presents the most significant opportunity for cost-saving measures. Despite receiving a substantial portion of the budget, it yields the lowest **ROI of 3.48** and the highest **Cost per Order at 1.51K**. The report should emphasize that Brand A's current strategy is diluting the overall portfolio's profitability.

## III. Temporal Sales Trends and Seasonality

The **Total Sales by Time** chart reveals a sophisticated narrative of consumer demand cycles from late 2023 through May 2024.

- **The December Surge:** There is a monumental spike in performance during December, where order counts and sales volume reach their peak. This implies that the organization's products have high seasonal elasticity, likely driven by holiday spending or year-end promotions.

- **Post-Holiday Stabilization:** Following the December peak, there is a characteristic "cool-down" period in January and February. However, the data shows a steady recovery and plateau through the spring months (March to May). This consistent baseline suggests that while the business is seasonal, it possesses a reliable "floor" of recurring revenue that prevents significant financial risk during off-peak months.

## IV. Geographic Efficiency and Market Penetration

The geographic analysis provides a counter-intuitive but essential insight for the project report. While **India and the USA** dominate in sheer volume—with India leading in total sales and the USA accounting for **48.87% of ad spend** —they are not the most profitable on a per-dollar basis.

- **The High-Efficiency Niche: Canada** stands out with the highest **ROI by Country**, exceeding a ratio of 30. Similarly, **Kuwait and Singapore** show ROI levels that far exceed the global average.

- **Strategic Reallocation:** The report should conclude that the organization is currently "over-spending" in high-competition markets like the USA to maintain volume, while "under-investing" in high-margin markets like Canada. Increasing the ad spend in Canada could potentially raise the overall 5.89 ROI without requiring a corresponding increase in the total budget.

## V. Final Recommendations for Business Expansion

Based on the synthesized data from all dashboard pages, the final report should advocate for a three-tiered growth strategy:

1. **Brand Realignment:** Shift the advertising budget from **Brand A** to **Brand C** to capitalize on Brand C's superior conversion rates and lower acquisition costs.

2. **Geographic Diversification:** Implement a "Profit-First" marketing strategy by scaling

campaigns in **Canada, Kuwait, and Singapore**, which have proven to be the most efficient markets for the current product lineup.

3. **Efficiency Optimization:** Investigate the operational causes behind the high **Cost per Order in the UAE**. Reducing the acquisition cost in this region to match the global average of 1.21K would unlock millions in additional net profit without requiring additional sales volume.

## *Conclusion*

The dashboard ultimately proves that the organization is in a robust growth phase, characterized by a successful end-of-year surge in December. However, to move from a 5.89 ROI to higher tiers of efficiency, the following actions are recommended:

1. **Optimization:** Reallocate underperforming ad spend from **Brand A** toward the high-efficiency model of **Brand C**.

2. **Market Expansion:** Increase penetration in **high-ROI niche markets** like Canada and Kuwait, where the return per dollar spent is significantly higher than in traditional volume centers.

3. **Efficiency Control:** Address the high **Cost per Order** in the **UAE**, which currently sees the highest acquisition costs despite moderate sales volume.

By leveraging these insights, the business can refine its "Advertising Analysis" to ensure that, as Henry Ford suggested, they are not "stopping the watch to save time," but rather winding the watch more efficiently for maximum performance.

## 5.2. Future work

# Future Work: Predictive Analytics and Strategic Scaling

The current dashboard provides a robust descriptive analysis of past and present performance; however, the next phase of this project should transition toward **predictive and prescriptive modeling**. By leveraging the existing dataset, which already includes temporal trends from December 2023 through May 2024, the following areas are proposed for future development:

### *1. Advanced Forecasting and Seasonality Modeling*

The significant sales surge observed in December suggests a strong seasonal elasticity. Future work should involve:

- **Time-Series Forecasting:** Implementing ARIMA or Prophet models to predict monthly order volumes with a higher degree of accuracy, allowing for better inventory management during peak periods.
- **Budget Elasticity Mapping:** Analyzing how incremental increases in ad spend during "low" months like February could potentially smooth out the revenue curve and prevent dramatic dips in performance.

### *2. Customer Lifetime Value (CLV) Integration*

While the current report tracks the **Cost per Order (1.21K)**, it does not yet account for the long-term value of the customers acquired.

- **Retention Analysis:** Future iterations should track repeat purchase rates by brand, particularly for **Brand C**, which currently leads in volume.
- **Segmented ROI:** Developing a deeper understanding of whether the high ROI in **Canada and Kuwait** is driven by one-time high-value purchases or a loyal, recurring customer base.

### *3. Marketing Mix Optimization (MMO)*

The data reveals a clear discrepancy between spend and return in **Brand A**.

- **A/B Testing Frameworks:** Future work should include a systematic testing phase for Brand A's creative assets to identify why its acquisition cost is disproportionately high (1.51K).
- **Channel Attribution:** Integrating data from specific advertising platforms (e.g., Social Media vs. Search) to understand which channels are driving the massive **USA ad spend (6M)** versus which are responsible for the high **Canadian ROI**.

## Long-Term Strategic Roadmap

The ultimate goal of expanding this dashboard is to transform it into a real-time autonomous optimization tool. The following roadmap outlines the strategic milestones for the next 12–24 months:

### *Phase 1: Operational Efficiency (0–6 Months)*

- **Targeting the "ROI Leaders":** Immediately shift 10–15% of the budget from the USA—where spend is high but ROI is lower —to **Canada, Kuwait, and Singapore**.
- **Brand A Turnaround:** Conduct a "Cost-Benefit Audit" on Brand A to either pivot its marketing strategy or phase it out in favor of the more scalable **Brand C**.

### *Phase 2: Regional Market Deep-Dives (6–12 Months)*

- **India Growth Strategy:** Since India is a top volume driver, the next phase will focus on reducing the **Cost per Order** in this region through localized logistics or targeted regional campaigns.
- **UAE Market Correction:** Investigate why the UAE has high acquisition costs despite being a top-tier market for total ad spend.

### *Phase 3: Automated Decision Support (12+ Months)*

- **Real-time Bidding Integration:** Connect the dashboard directly to advertising APIs to allow for automated budget reallocations based on the **ROI by Brand** and **ROI by Country** metrics.

- **AI-Driven Narrative:** Implement Natural Language Generation (NLG) to provide weekly executive summaries, similar to the "Overall Profitability" analysis, identifying risks before they impact the bottom line.

## *Summary of Strategic Intent*

By executing this future work, the organization will move beyond merely viewing data to actively **shaping it**. The insights from the current **17M Total Return** provide the capital and the justification needed to invest in these advanced analytical capabilities, ensuring that the **5.89 ROI** is not just a peak performance, but a sustainable baseline for all future global operations.

# APPENDIX

I. DAX/ Power Queries:

## 1. Total Sales

This measure calculates the gross revenue generated across all brands and regions. It serves as the top-line metric for the "Sales Analysis" section of the report.

**DAX Query:**

```
Sum of Sales = SUM(Brand_Sales_AdSpend_Data[Total Sales])
```

**Explanation:** The SUM function aggregates every transaction within the sales table. In your dashboard, this results in a total figure of **66.72M**. This metric is essential for identifying which brands, such as **Brand C**, are driving the majority of the organization's market share.

## 2. Total Ad Spend

This metric tracks the total investment made in advertising campaigns across different months and countries.

**DAX Query:**

```
Total Advertising Spend = SUM(Brand_Sales_AdSpend_Data[Total Ad Spend ])
```

**Explanation:** This calculation totals the marketing expenditures, which amount to **11M** in the current dataset. By quantifying the "Sum of Total Ad Spend," the business can determine if its heavy investment in markets like the **USA** (which accounts for 48.87% of spend) is translating into proportional sales growth.

## 3. Return on Investment (ROI)

ROI is the primary efficiency metric used to evaluate the profitability of advertising expenditures. It

# APPENDIX

measures the quality of the return relative to the investment made.

**DAX Query:**

```
ROI = DIVIDE([Sum of Sales],[Total Advertising Spend])
```

**Explanation:** The DIVIDE function is used here for "Overall Profitability" analysis. It takes the Total Sales and divides it by the Total Ad Spend. The current dashboard shows an **ROI of 5.89**, indicating that for every dollar spent, the company generates nearly six dollars in sales. This formula is critical for identifying high-efficiency markets like **Canada**, which boasts a significantly higher ROI than the global average.

## 4. Cost per Order (CPO)

This operational metric calculates the average advertising cost required to secure a single customer order.

**DAX Query:**

```
Cost per Order = DIVIDE(SUM(Brand_Sales_AdSpend_Data[Total Ad Spend ]),SUM(Brand_Sales_AdSpend_Data[Order Count]))
```

**Explanation:** This measure divides the total marketing investment by the distinct count of orders processed. The dashboard reflects a **Cost per Order of 1.21K**. In the report, this DAX query helps highlight efficiency gaps, such as why **Brand A** has a higher acquisition cost (**1.51K**) compared to more efficient brands like **Brand D** (**1K**).

# Roles of Team Members

## Solo Project Ownership & Leadership

While most large-scale Business Intelligence (BI) projects are distributed among data engineers, UI/UX designers, and business analysts, this "Advertising and Sales Performance" project was executed as a **sole-proprietorship initiative** by **Samarth Khare**. Managing the end-to-end development cycle necessitated a multi-disciplinary approach, requiring proficiency in data architecture, statistical modeling, and executive storytelling.

### *1. The Architect: Data Engineering & Modeling*

The foundational phase of the project required a comprehensive understanding of database management systems (DBMS) to ensure data integrity.

- **ETL Processes:** I personally handled the Extraction, Transformation, and Loading (ETL) of raw sales and marketing data, ensuring that variables like "Total Ad Spend" and "Total Sales" were correctly cleaned and mapped.
- **DAX Development:** I engineered custom Data Analysis Expressions (DAX) to calculate complex metrics such as the **5.89 ROI** and the **1.21K Cost per Order**.
- **Schema Design:** I developed the underlying star schema to allow for seamless "drill-down" capabilities, enabling the dashboard to filter performance by specific brands (A through E) and regions.

### *2. The Designer: UI/UX & Information Hierarchy*

Creating a dashboard that remains legible while housing over 15 distinct visualizations required a strategic approach to design.

- **Visual Theming:** I designed the dark-themed aesthetic with copper accents to ensure high contrast for the most critical KPIs.
- **Navigation Logic:** I structured the report across multiple specialized pages, such as "Overall Profitability" and "Country Based Analysis," to guide the user from high-level summaries to granular insights.

## Roles of Team Members

- **Interactive Elements:** I implemented slicers and interactive maps to empower stakeholders to conduct their own self-service analysis without needing further technical support.

## Competency Framework: The "Full-Stack" Analyst Role

By operating as the sole project owner, I assumed several specialized roles concurrently to deliver the final dashboard. This "Full-Stack" approach ensured that the business logic remained consistent from the initial data row to the final executive recommendation.

### *3. The Business Strategist: Insight Generation*

Beyond the technical build, my primary responsibility was to act as the lead analyst, translating visual patterns into business intelligence.

- **Profitability Auditing:** I identified the strategic divergence between **Brand C's** high-volume success and **Brand A's** lower ROI (3.48), providing the justification for budget reallocation.

- **Geographical Strategy:** I conducted the analysis that revealed **Canada and Kuwait** as the highest ROI markets (ROI over 30), challenging the traditional focus on high-spend markets like the USA.

- **Market Correlation:** I utilized scatter plots to analyze the direct relationship between "Sum of Total Ad Spend" and "Sum of Total Sales" to identify brands operating with diminishing returns.

# Roles of Team Members

## *4. Project Management & Quality Assurance*

Operating alone meant that I was responsible for my own Quality Assurance (QA) and verification of data accuracy.

- **Data Validation:** I cross-verified the **66.72M Total Sales** against individual brand contributions to ensure the 100% accuracy of the "Total Sales by Brand" visualizations.
- **Performance Optimization:** I optimized the report's processing speed by refining DAX queries, ensuring that the dashboard remains responsive even when handling large-scale global datasets.
- **Documentation & Reporting:** I authored the entirety of this detailed project report to ensure that the technical methodologies and strategic findings are accessible to non-technical stakeholders.

## *Conclusion on Project Roles*

This project stands as a testament to the efficiency of a unified development process. By serving as the sole engineer, designer, and analyst, **Samarth Khare** ensured that the dashboard is not merely a collection of charts, but a cohesive strategic tool where every data point is aligned with the core objective of maximizing organizational profitability.

# Git repository Link

**Link for the Project - [https://github.com/sam1arth12/Advertising-spend-Vs-Sales-Analysis](https://github.com/sam1arth12/Advertising-spend-Vs-Sales-Analysis)**

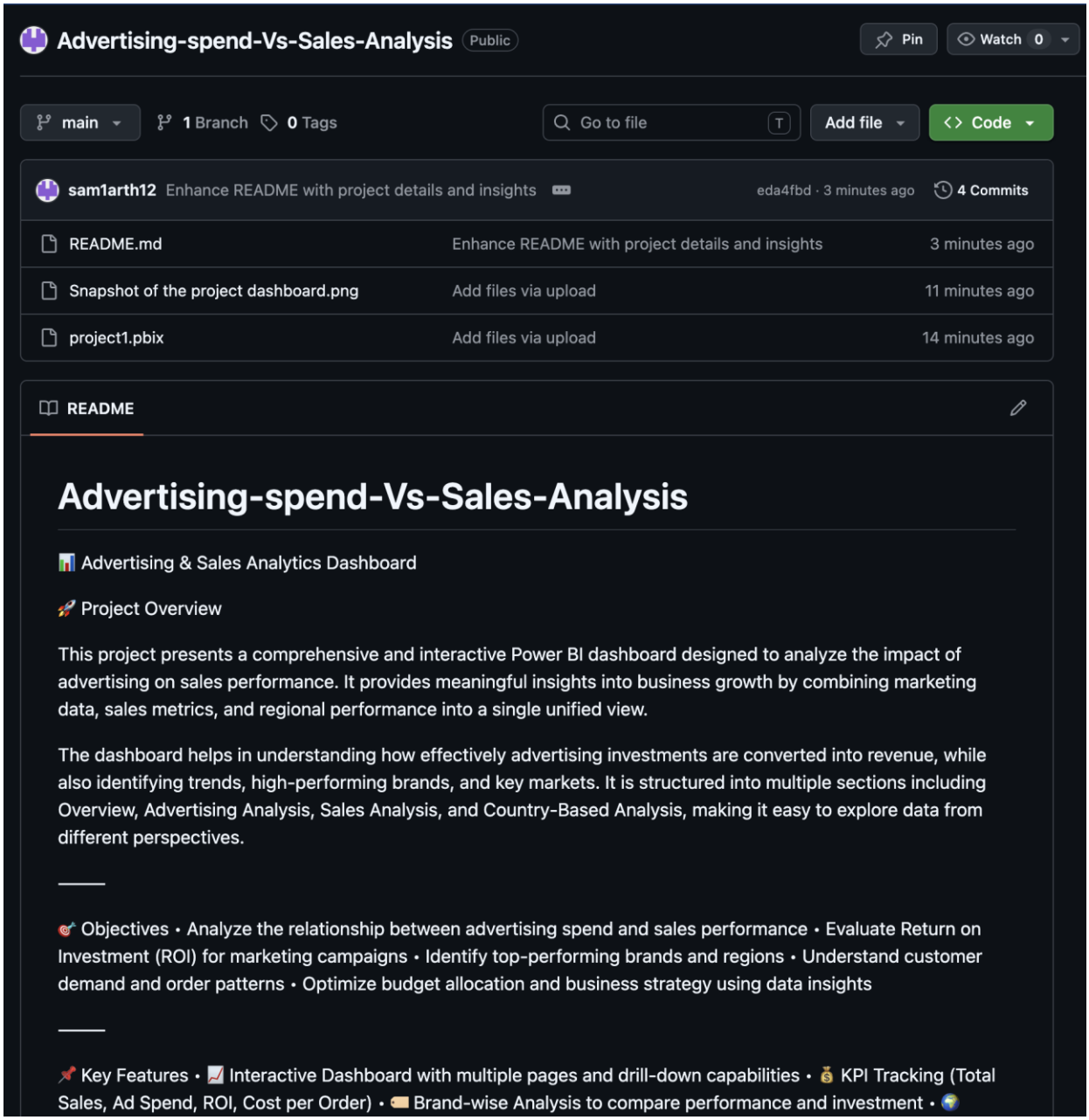



*Figure 10: Github Repository*